

Online Neuroevolution as an Ensemble Generator: Island-Champion Aggregation for Cost-Aware Stock Return Prediction

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Working draft — 2026-08-29

Abstract

Online neuroevolutionary architecture search (ONE-NAS) maintains a population of small recurrent networks that evolves continuously as a data stream arrives. Its published prediction rule — forecast with the single best genome — presumes that the search can identify its best network. On equity return panels it cannot: validation fitness is nearly uncorrelated with realized economic quality, and the single champion underperforms a plain online LSTM. We show that the population is nevertheless valuable — as an *ensemble generator*. Aggregating each island’s champion by rank-mean turns the same completed runs into the strongest system in a cost-aware prequential evaluation on CRSP mid-cap panels: within-run, the ensemble beats the single champion by +23.0 net points over 2022–2024 (paired $t=18.2$ over ten seeds), a gain that widens with island count, replicates on a disjoint 2016–2019 span, is positive in all fifteen hyperparameter configurations we trained, and matches the ensemble-averaging prediction from measured member diversity to within 4%. The resulting system exceeds online LSTM/GRU and periodic-retraining baselines under identical books and costs, is the only arm with significant factor-model alpha (+7.9%/yr, Newey–West $t=2.14$), and its always-online population appreciates: a cold start forfeits roughly half the profit. The networks are tiny (tens of weights per member), making the entire system — search included — deployable on commodity single-board hardware.

1 Introduction

Time-series forecasting models are typically compared on pointwise error, yet a lower forecast error need not imply a better trading decision. The prior work in this line compared offline-evolved recurrent networks (EXAMM) against modern forecasters under a daily long–short protocol and found the evolved networks strongest. This paper studies the *online* variant, ONE-NAS, under a continuously evolving population, and makes four contributions:

1. **A prediction rule for online NAS:** rank-mean aggregation of island champions, replacing the published single-genome rule.
2. **A cost-aware prequential protocol** on CRSP mid-cap panels: pooled cross-sectional training, netted half-spread costs, paired seed-level statistics, and a pre-registered configuration ledger.
3. **A mechanism study** showing *why* aggregation is the right rule: fitness is uninformative for selection on this domain, so the population’s value is only accessible by averaging — and the gain matches diversity theory quantitatively.

4. **An online-continuity result:** the deployed population is an appreciating asset; cold-starting forfeits about half the profit.

2 Method

ONE-NAS extends EXAMM neuroevolution to streaming data: islands of small recurrent genomes (evolvable node types: simple, UGRNN, MGU, GRU, Δ -RNN, LSTM; deep recurrent connections with time-skips) train online as each new window of the stream arrives, with island extinction and repopulation maintaining structural diversity. The published system predicts with the single global-best genome. We instead form, at every generation, the **rank-mean ensemble of the island champions**: each island’s best genome produces a cross-sectional prediction; predictions are rank-transformed per day and averaged. Islands evolve in isolation and exchange genetic material only through rare inter-island crossover, so champions are structurally diverse; their measured pairwise prediction correlation is $\rho \approx 0.17\text{--}0.22$, the diversity that aggregation harvests.

3 Data and evaluation protocol

Panels. Four 50-stock mid-cap panels (V1–V4) drawn from CRSP, identical to the prior paper’s validation portfolios. Inputs are seven per-stock features (return, cross-sectional rank input, bid–ask spread, Amihud illiquidity, 21-day reversal, turnover ratio, 21-day volatility); the target is the next-day cross-sectional rank-normal return. Feature normalization is frozen at the 2019 burn-in boundary; realized returns for all economics are CRSP total returns (split- and dividend-adjusted).

Clock. Prequential: sliding windows of length 40 advance 5 trading days per generation; no prediction uses information after its date. The population evolves from 2020-01 onward; headline evaluation uses 2022–2024, the prior paper’s test years, with the full 2020–2024 span in the appendix.

Book. Predictions trade through overlapping-sleeve top-10/bottom-10 long–short portfolios (Jegadeesh–Titman construction, 10-day holding, 1/10 of capital per day), charged the half quoted spread on netted traded notional. The book reads only the cross-sectional *order* of predictions and is invariant to monotone rescaling.

Statistics. The comparison unit is one training run. Ten seeds (42–51) per configuration; metrics are averaged over panels within seed and tested by paired t across seeds; daily quantities use HAC standard errors. The system configuration was frozen in a pre-registered ledger before confirmatory scoring; hyperparameters were selected on 2016–2019 data only (Section 7).

4 Headline results

Table 1 compares ONE-NAS on the evaluation window against the prior paper’s published EXAMM and buy-and-hold rows, under three portfolio constructions. Figure 1 shows the cumulative net paths. ONE-NAS is profitable in every trade year including the 2022 drawdown, and is the only arm whose factor-model alpha is significant: regressing pooled daily book returns on the Fama–French three factors plus momentum (Newey–West lag 10) gives $+7.9\%/yr$ ($t=2.14$, market beta 0.12) for the 40-island system, against $t \leq 1.8$ for every baseline and a beta-0.97 buy-and-hold control at $t=0.4$.

Table 1: Net return (%) by trade year, mean over $V1-V4$. ONE-NAS books are restarted at each window boundary; the 2022–24 column is one book run over the three-year window.

	2022	2023	2024	2022–24
ONE-NAS (40 isl., sleeves book)	+6.1±0.8	+12.4±0.7	+6.4±0.5	+27.5±1.1
ONE-NAS (40 isl., banded book) ^a	+12.5±1.2	+14.6±1.3	+3.4±0.9	+33.9±1.8
ONE-NAS (40 isl., Algorithm 1 book) ^b	+1.3±1.6	+12.8±1.0	+10.8±1.2	+25.0±1.8
ONE-NAS single champion (8 isl., sleeves) ^c	+3.1±1.8	+1.7±1.3	+2.7±1.1	+8.5
Online LSTM (sleeves book)	+1.2±1.3	+4.5±1.2	+3.8±0.7	+11.3±1.8
EXAMM (prior paper)	+15.5	+9.1	+8.5	+33.1 ^d
Buy & hold (prior paper)	−10.1	+11.2	+6.8	+7.9 ^d

^a Buy/hold-spread rule: enter at rank ≤ 10 per side, exit when rank decays past 35 (Novy-Marx & Velikov 2016). Winner of a seven-book sensitivity sweep on the development span; reported as sensitivity, not an adopted headline.

^b The prior paper’s conditional long–short book applied verbatim. On rank-normal predictions its entry gate fires every day (average holding 1 day vs. EXAMM’s ≈ 5), an $\approx 8\%$ /yr cost drag; conservative for ONE-NAS.

^c The published ONE-NAS prediction rule; seeds 42–44 ($n=3$).

^d Prior-paper rows are its published Table 7 yearly cells from its own trading stack; their 2022–24 column is the sum of yearly cells.

Table 2: Prediction rule, within-run paired: single global-best genome vs. island-champion rank-mean ensemble from the same runs. Net % / Sharpe; Δ = ensemble – single, paired t over seeds.

Fleet	Window	Single	Ensemble	$\Delta_{\text{net}} (t)$	$\Delta_{\text{Sharpe}} (t)$
40 isl. ($n=10$)	2022–24	+4.5 / 0.21	+27.5 / 0.87	+23.0 (18.2)	+0.66 (14.7)
40 isl. ($n=10$)	2020–24	+2.7 / 0.08	+45.6 / 0.68	+42.9 (14.6)	+0.60 (13.4)
8 isl. ($n=10$)	2022–24	+8.3 / 0.33	+20.5 / 0.70	+12.2 (6.7)	+0.37 (5.8)
8 isl. ($n=10$)	2020–24	+12.0 / 0.21	+32.1 / 0.52	+20.1 (5.5)	+0.30 (5.3)
8 isl. ($n=5$) ^a	2016–19	+16.3 / 0.56	+26.0 / 0.92	+9.7 (5.2)	+0.36 (4.5)
40 isl. ($n=5$) ^a	2016–19	+8.9 / 0.30	+36.7 / 1.22	+27.8 (7.5)	+0.92 (6.6)

^a Robustness replication on the 2016–2019 tuning-span fleets.

5 Why it works: aggregation, not selection

The fitness signal cannot select. The rank correlation between a genome’s validation MSE and its realized next-week economic quality is +0.02; given a fixed pool ten times the usual size, selecting the argmax buys +0.022 daily rank IC while simply averaging the pool buys +0.143; a random pick beats the argmax. The identical harness passes its positive control on NAS-Bench-201, so the limitation is the domain’s, not the tooling’s: at this signal-to-noise, per-genome quality is unselectable.

Within-run test. Table 2 extracts both prediction rules from the same completed runs. The ensemble wins at every fleet, window, and metric (t from 4.5 to 18.2); the single champion *degrades* as islands widen (more lottery tickets for an uninformative selector) while the ensemble improves — the winner’s curse and its cure in one table. The contrast replicates on 2016–2019, a span that played no role in adopting the ensemble rule.

The gain is quantitatively explained by diversity. With measured member rank IC m

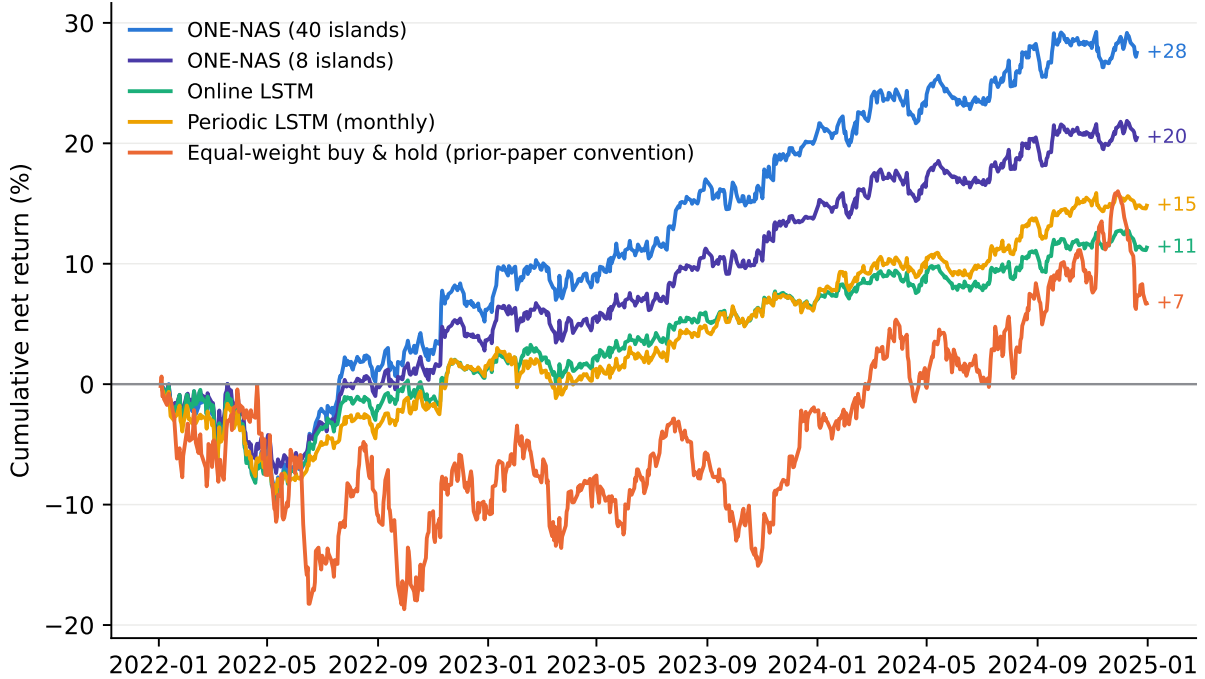


Figure 1: Cumulative net return, 2022–2024 (books restarted at the window start; endpoints match Table 1). Buy-and-hold follows the prior paper’s benchmark convention.

and pairwise prediction correlation ρ , the equicorrelated averaging prediction $m\sqrt{N/(1+(N-1)\rho)}$ matches the measured ensemble IC within 4% in all four fleets (e.g. 40 islands: predicted +0.0132, measured +0.0128 at $\rho=0.172$). Wider populations are more diverse (ρ 0.216 at 8 islands vs. 0.172 at 40), which is where width’s ensemble gain originates; width otherwise buys reliability (seed dispersion of Sharpe falls from 0.171 to 0.078), not mean IC.

The gain is configuration-invariant. Re-scoring both prediction rules inside all fifteen hyperparameter configurations we ever trained (Section 7) gives a positive within-run ensembling net gain in 15/15 cells (mean +8.2, up to +20.6, $t=6.0$); it shrinks toward zero only in the two IC-selection cells whose populations are degenerate — aggregation harvests diversity from healthy populations.

Boundary. Random-architecture online ensembles match evolved champions at the registered book ($t=0.41$): on this domain, evolution’s contribution beyond generating diverse small architectures is not detectable, and aggregation doubles Sharpe across all member-generation mechanisms. We state this as a scope condition on the method’s value.

6 Online continuity

Deploying the frozen configuration cold at 2022-01 earns +12.3 pooled net over 2022–2024 versus +26.5 for the same window inside runs that had been evolving online since 2020 — the always-online population is an appreciating asset, and this is what online search buys over periodic retraining (a generation-count confound is disclosed in the artifact).

Table 3: ONE-NAS hyperparameter settings.

Hyperparameter	Value
Islands	40 (headline); 8 (registered primary)
Elite population per island	8
Generated genomes per island per generation	5
Repopulation frequency	every 50 generations
Selection metric	validation MSE
Prediction rule	rank-mean ensemble of island champions
Target	1-day cross-sectional rank-normal return
Sequence length / window step / validation sets	40 / 5 / 5
Training subsequences per genome	2000
Backpropagation epochs per genome	10
Replay (PER) α , λ , ϵ	0.6, 0.007, 10^{-8}
Rounds per generation	1
Node-type library	simple, UGRNN, MGU, GRU, Δ -RNN, LSTM
Mutation / intra- / inter-island crossover	0.7 / 0.2 / 0.1 (EXAMM defaults)
Warm-up generations	50

7 Hyperparameters

ONE-NAS hyperparameters were selected by single-factor search scored on 2016–2019 validation data, disjoint from the 2020–2024 evaluation span; the search covered island count (8–40), elite and generated population sizes (including the originally published 5/10), repopulation frequency (25–100), training-subsequence count (600–2000), sequence length (15, 40), backpropagation epochs (10, 20), replay-priority parameters, rounds per generation, selection metric (validation MSE vs. rank-IC variants), and target horizon (1- and 5-day cross-sectional). The selected settings are given in Table 3; complete per-configuration results are available in the project repository. Two findings are noteworthy: selecting genomes by validation rank-IC *harms* economics (−0.55 to −0.90 Sharpe), consistent with Section 5’s unselectability result; and no variant beat the selected configuration on the tuning objective.

8 Deployment

Each ensemble member averages tens of weights; even the 40-champion ensemble totals $\approx 2,600$ parameters, orders of magnitude below the transformer baselines of the prior study. The search itself is CPU-only with a weekly decision clock, giving large headroom for on-device operation; on-device measurements (ensemble inference and one weekly evolution generation on single-board hardware) and live-pilot status are reported in the final version. [D1/D2 measurements and pilot paragraph to be inserted.]

9 Limitations

The evaluation window (2022–2024) is inherited from the prior paper’s protocol and lies within this project’s development span; full-span results and the configuration ledger are provided for transparency. The universe carries survivorship bias from drawing current constituents backward.

Results cover one market, one frequency, and mid-cap panels; the random-architecture boundary result limits claims about evolutionary search per se, as opposed to the online-population-plus-aggregation system. Prior-paper comparison rows follow that paper’s published benchmark convention, noted where used.